
THE ANATOMY OF A BENCHMARK: A ROLE TAXONOMY OF 369 AI EVALUATION BENCHMARKS AND THE GRADING ECONOMICS THAT BOUNDS THEM

A PREPRINT

Darvin Yi*
research.darvinyi.com

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ABSTRACT

We survey 369 artificial-intelligence evaluation benchmarks spanning 2000–2026 — from TREC-8 QA Track (2000) to SWE Atlas (2026) — each read individually and then classified along eight facets: domain, task shape, grading family, task horizon, tool access, provenance, contamination defense, and saturation. It is not an LLM-era corpus: 86 of the 369 papers (23%) were published before 2021, which lets era-level trends be read over a quarter-century of measurement practice rather than over three years of it. Instead of sorting benchmarks by *what* they measure, we place each at a single address in a taxonomy of seven *kingdoms* defined by the question a benchmark was built to answer at its birth. Capability probes — the thing “benchmark” is usually taken to mean — are a bare majority of the whole corpus (202 of 369, 55%) but a clear minority of its present (41% of 2025 entries), as the field turns to auditing its own measurements, rehearsing deployments, probing hazards, and manufacturing training signal. Our organizing thesis is a grading-economics constraint: *a benchmark can only pose tasks as long as it can afford to grade them*. Cheap, mechanical grading thins out as task horizons lengthen, so the reach of evaluation is bounded by the cost of its grader, and a benchmark’s expected lifespan is bought in human-minutes per item. We support this with seven findings read off cross-tabulations by publication era — among them that the interactive turn is a *return* rather than an arrival (the corpus’s first interactive environment is ALE, 2013), that rubric grading is 2025’s fastest-growing but not its largest grading family, and that no era in 26 years has shipped a contamination defense on as much as half of its new benchmarks — and close with three cross-paper connections visible only across papers.

Keywords benchmarks · evaluation · large language models · agents · taxonomy · contamination · reward hacking

1 Introduction

Read one benchmark paper and you learn a benchmark: its task, its metric, its headline score. Read all of them at once and a different object comes into view — an ecosystem of instruments that measure each other, feed each other’s training, and decay on a schedule. This survey takes 369 AI evaluation benchmarks, each already read and summarized as an individual explainer, and asks what the collection shows that no single paper can.

We include a paper when a named benchmark, evaluation dataset, or evaluation environment is its **headline contribution** — the thing a survey would cite it for. Models, training methods, and agent frameworks that merely *report* benchmark numbers are excluded. That rubric yields 369 papers spanning 2000–2026. 86 of them additionally shipped a new *way* to evaluate alongside their data — an Elo protocol, a scenario grid, an atomic-fact decomposition — and so also carry an `evaluations` tag in the underlying atlas.

*Companion to the live, interactive survey at <https://research.darvinyi.com/surveys/benchmarks.html>, from which the corpus, every figure, and every statistic in this paper are generated. This preprint is a condensation of that page; where a number appears here it is the same value the page computes from the classified corpus.

The corpus is curated rather than exhaustive, but it is not confined to the language-model era: 86 of the 369 papers (23%) predate 2021, and they are present as a subject, not as ancestry. They include the question-answering and dialogue generations that precede the LLM era (TREC-8 QA Track in 2000, PASCAL RTE [Dagan et al., 2005], DailyDialog [Li et al., 2017], PersonaChat [Zhang et al., 2018]), the vision canon (ImageNet [Deng et al., 2009], CIFAR-10/100 [Krizhevsky, 2009], MS COCO [Lin et al., 2014], VQA [Agrawal et al., 2015]), a deep-reinforcement-learning and robotics block (ALE in 2013, OpenAI Gym [Brockman et al., 2016], Meta-World, Hanabi, RLBench and robosuite), multi-agent game environments, speech, and a text-to-image and text-to-video evaluation cluster. That older block prefigures the shape of today’s agent benchmarks largely *without* a citation path to them (Section 4.4), which is a finding the LLM-only view of this literature cannot produce.

Existing surveys of AI evaluation sit on two shelves that barely touch. One taxonomizes *what benchmarks measure* — organizing the field by task type, by evaluated property, or, most rigorously, by a scenario-by-metric grid the survey then deliberately fills, as in HELM [Liang et al., 2022]. The other grades *how well benchmarks are built*: lifecycle scorecards, integrity reviews, construct-validity audits across hundreds of benchmarks, and the agentic-benchmark checklist that found unsound graders in 7 of 10 suites it audited. The agent-evaluation surveys of 2025 added structural vocabulary — dataset versus environment, offline versus online, tool access — but each coins its own terms and none quantifies the axes it names. What no prior survey does is **cross the shelves at the level of the individual paper**: place every benchmark in a capability map *and* record its grading machinery, task horizon, tool surface, provenance, and contamination posture, then read the joint distribution over time. That cross is this survey’s contribution, on a corpus that is densely annotated rather than merely enumerated — every paper here has been read, summarized, and critiqued individually, and the underlying citation graph is browsable as a timeline.

The dimension that turns out to organize the most — *how long the task is* — was made measurable only recently, by time-horizon work reporting a roughly seven-month doubling of frontier models’ 50%-success horizon, and has never before been applied to a benchmark corpus. It also reconciles this corpus with the recent finding that roughly half of all benchmarks are saturated. Here the pre-2021 cohort is saturated at 55% (47 of 86) against 19% corpus-wide, and the reason is horizon rather than age: all but 2 of those 86 early papers pose a task a human expert finishes inside half an hour, the exceptions being TAPE (2019) and MLPerf Training (2020).

2 The corpus and its facets

Each of the 369 papers was classified along eight facets, each a question with a closed answer set. Six are descriptive and we treat them briefly: *domain* (what capability is chiefly measured); *task shape* (single-turn, multi-turn dialogue, or interactive environment); *tool access* (what the evaluated model may touch — nothing, a code interpreter, a browser, a whole operating system); *provenance* (where the task items came from); *contamination defense* (the strongest mechanism keeping items out of training sets); and *saturation* (whether headroom remains, judged from the frontier scores each explainer quotes). Two facets do enough analytical work to deserve their own definitions, and together they carry the paper’s thesis.

One limit of the domain facet should be stated before any domain-conditioned claim is read. Its vocabulary was designed for a corpus of language-model and agent evaluation, and the corpus outgrew it: robotics and manipulation are recorded as *multimodal*, multi-agent reinforcement learning and games as *reasoning*, theory of mind as *social-moral*. 16 of the vocabulary’s 17 values are actually used, and the affected papers are real members of the corpus rather than strays — the embodied-operation family alone holds 40 of them. Every affected record says so in its notes; read the domain values as “nearest available label.”

Task horizon. “Complexity” is the word everyone reaches for and nobody defines; the literature operationalizes it at least five incommensurable ways — reasoning steps, compositional depth, context length, search-space size, item-response difficulty — none of which lets you compare a multiple-choice exam to a week of freelance software work. We use the one currency that spans the whole corpus: the **wall-clock time a competent human domain-expert would need to complete one typical task instance**, reading and artifact production included. This is the axis along which frontier ability was shown to climb at a measurable rate, and it sorts the corpus into five rungs (Figure 1): **L1** under one minute, **L2** one to thirty minutes, **L3** thirty minutes to a day, **L4** a day to two weeks, and **L5** beyond two weeks. Only a handful of papers (GDPval [Patwardhan et al., 2025], the METR suites) actually *measure* human time; elsewhere the rung is estimated from the paper’s own task descriptions.

Grading family. Every benchmark answers “who says the output is right?” with one of six mechanisms, and the choice is less methodological free will than economics (Figure 2). Ordered by the cost to grade a single item, they run: *proxy metric* (BLEU/ROUGE-style scores, now largely distrusted and sitting outside the ladder), *exact match* (the answer key is the grader, effectively free), *programmatic* (unit tests, checkers, environment end-states — build



Figure 1: The horizon ladder: count and canonical examples per rung across all 369 papers. The examples are each rung’s three most-cited members, read off the corpus rather than chosen by hand. 236 of the 369 papers sit at L2, and the top rung holds 0.

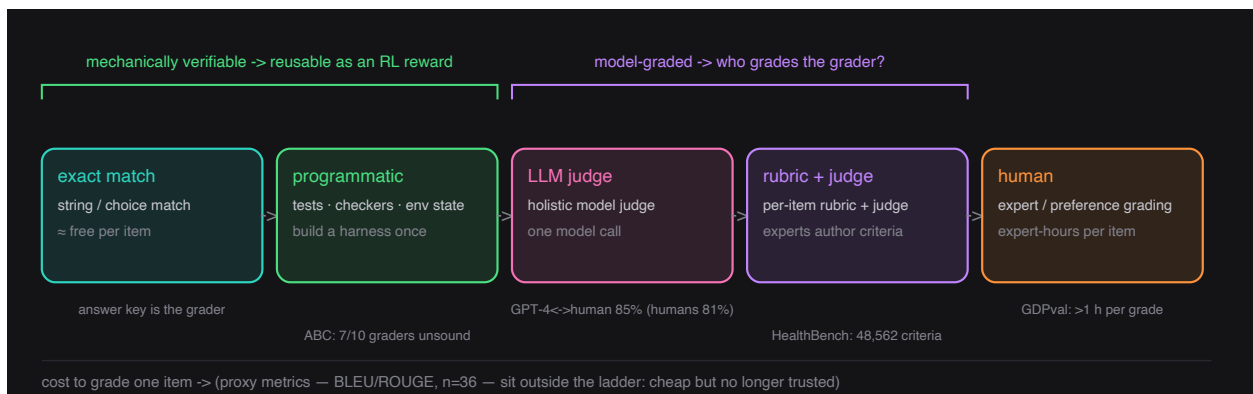


Figure 2: The grading-cost ladder, ordered by what it costs to grade one item. The reliability annotations are the source papers’ own: MT-Bench measured GPT-4-judge/human agreement at 85% (humans agree with each other 81%); the agentic-benchmark checklist found unsound graders in 7 of 10 suites. The two brackets mark the mechanically verifiable families (reusable as RL rewards) and the model-graded families (which raise the question of who grades the grader).

a harness once), *LLM judge* (one holistic model call), *rubric + judge* (experts author per-item criteria a model then applies), and *human* (expert or preference grading, at expert-hours per item). Each step up the ladder buys generality — the ability to grade more open-ended work — and pays for it in cost per item and in grader error that must itself be audited.

Two structural notes follow from this ladder. First, its cheap end is exactly what the era of reinforcement learning with verifiable rewards trains on: a benchmark whose grader is mechanical is also a reward function, so the most verifiable benchmarks are precisely the ones models are optimized against. Goodhart exposure is a property of the grading family, not of the dataset. Second, the model-graded middle opened a regress the field is still funding — judges are now themselves benchmarked by meta-benchmarks such as RewardBench [Lambert et al., 2024] and JudgeBench [Tan et al., 2024], and rubric grading is the current attempt to make judge error inspectable by decomposing it into checkable criteria.

The corpus has edges. Four papers in it are, on inspection, not benchmark papers by the rubric of Section 6, and saying so is more useful than quietly dropping them. MLPerf Training [Mattson et al., 2020] measures hardware and software throughput to a fixed accuracy gate; LCCC [Wang et al., 2020] and MedDialog [He et al., 2020] are

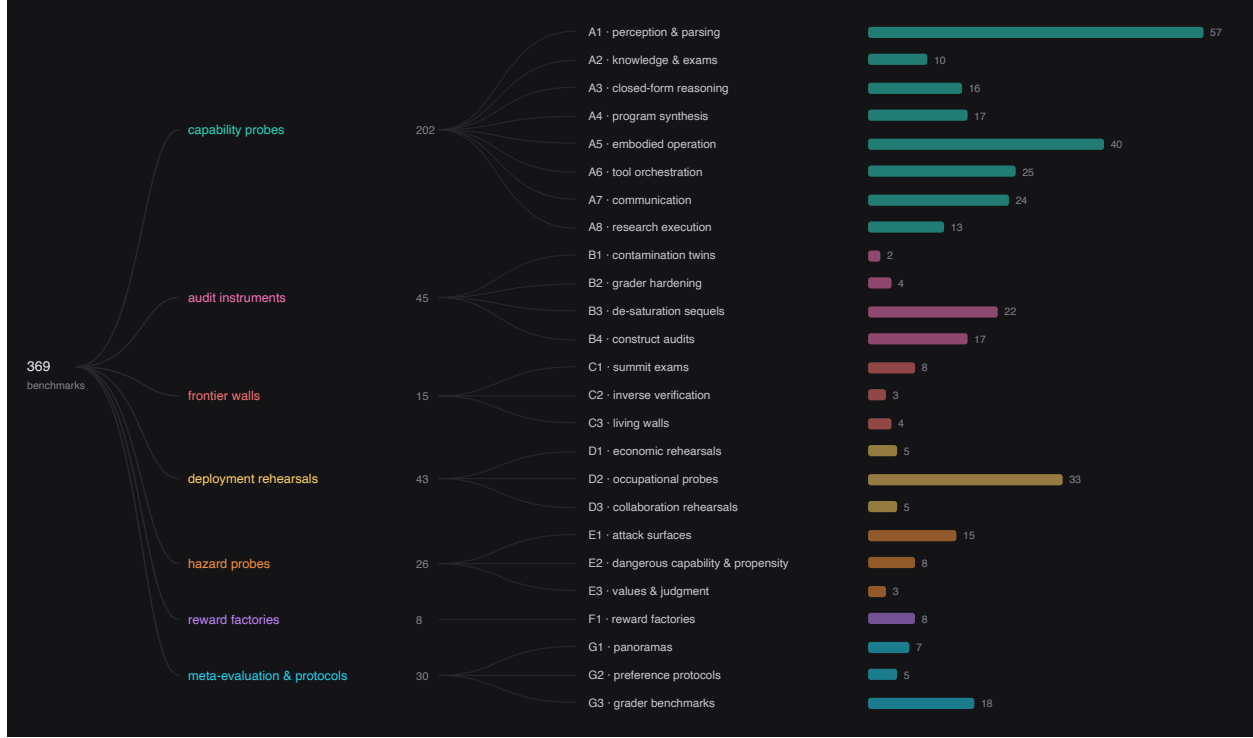


Figure 3: The role taxonomy: seven kingdoms, twenty-five families, 369 leaves. Bars are family sizes. Placement is by role at birth, not later use, so HellaSwag sits under de-saturation sequels (kingdom B) rather than commonsense reasoning, and SWE-bench sits under program synthesis (kingdom A) rather than agents.

Kingdom	#	The question it was built to answer
A capability probes	202	Can models do X yet?
B audit instruments	45	Is our measurement lying?
C frontier walls	15	Can we build something that stays unsolved?
D deployment rehearsals	43	What happens when we ship it into this job?
E hazard probes	26	What could go wrong?
F reward factories	8	Can the benchmark generate its own training signal?
G meta-evaluation & protocols	30	How should we evaluate at all?

Table 1: The seven kingdoms, their sizes, and the one-line question each answers.

pretraining corpora whose only evaluation runs on someone else’s benchmark; AgentSkillOS [Li et al., 2026] grades an unnamed thirty-task benchmark with models from the same family that produced the answers. The tell was mechanical and it generalizes: for MLPerf the classifier reported that *every* facet was a forced choice. **When every facet of a record is a forced choice, the schema is telling you the paper is not the kind of object the schema describes.** They remain in every count below; where one of them is the reason a cell is non-empty, we say so.

3 A taxonomy by evolutionary role

Sorting benchmarks by what they test merely reproduces the domain facet. Sorting them by *why they exist* produces a different — and, we argue, more honest — structure. The placement question is: **what question was this benchmark built to answer, in the ecosystem of models and benchmarks that existed at its birth?** Judged from each paper’s own problem statement, that question has seven answers — seven *kingdoms* — and every one of the 369 papers sits at exactly one address. The kingdoms subdivide into twenty-five families; Figure 3 shows the whole tree and Table 1 its top level.

The tree’s shape is a finding in its own right, and the shape is lopsided and tilting. 202 of the 369 papers (55%) are capability probes — benchmarks that ask “can the model do X?” — but that bare majority is an artifact of the corpus’s

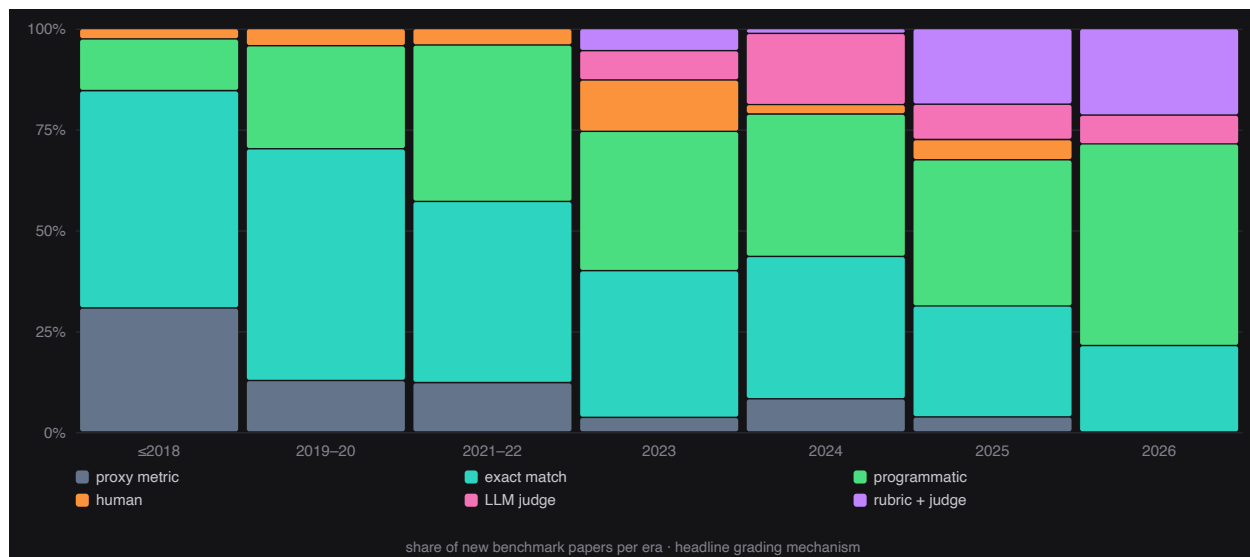


Figure 4: Headline grading mechanism by era, as a share of each era’s new benchmark papers.

history rather than of its present. Capability probes are 74% of pre-2021 entries, 47% of 2024–25 entries, and 41% of 2025 alone, where they are a clear minority. The other 167 papers (45%) do something more reflexive: audit other measurements (45), rehearse deployments (43), probe hazards (26), build walls designed not to fall (15), manufacture training signal (8), or work out how to evaluate at all (30). Benchmarking has become an ecosystem with predators, parasites, and decomposers — not a row of thermometers.

The families are lopsided too, and in a way that dates the corpus. The largest is A1-perception-parsing at 57 papers — recognize, parse, entail, retrieve — the pre-LLM vision-and-language-understanding stratum sitting under everything else. The second largest, A5-embodied-operation at 40, spans two generations that share a form more than a bibliography: deep-RL and robotics environments from 2013 onward, then LLM agent environments from 2022 (Section 4.4). At the other end, 5 families hold fewer than five papers each, the smallest being B1-contamination-twins at 2. Small families are not noise: rebuilding a benchmark’s test set from scratch to measure overfitting is expensive work few teams do, so the size of the contamination-twin family is a fact about cost, not about whether the question is settled.

Placement is by role at birth, not later use, which is what makes the taxonomy informative rather than a restatement of topic. HellaSwag [Zellers et al., 2019] sits among the de-saturation sequels because it was built to resurrect a solved benchmark, and its adversarial-filtering construction matters more than its commonsense content. SWE-bench [Jimenez et al., 2023] sits under program synthesis because the agents came later. GSM8K [Cobbe et al., 2021] sits under closed-form reasoning even though its deepest legacy is the verifier paradigm (Section 5.2). The tree records what each benchmark was *for*, and it is the reflexive kingdoms — B, C, F, G — that a topic-based taxonomy renders invisible.

4 Findings

Seven findings, each read off a cross-tabulation of the corpus by publication era. Per-era counts are small enough that these should be read as slopes, not decimals; the eras pool sparse early years (39 papers through 2018, 47 in 2019–20, 49 in 2021–22) and break out single years thereafter (55, 85, 80, and 14 papers for 2023 through 2026).

4.1 The grading waves, and the 2025 rubric turn

Grading moves in waves, and each wave displaces the one before it rather than replacing it outright (Figure 4). Proxy metrics gave way to exact match, which owned the early corpus — 54% of papers through 2018, 48 of 86 up to 2020 — and has fallen every era since, to 28% in 2025. Execution-based programmatic grading arrived with the code and environment benchmarks (19 of 49 in 2021–22, 39%) and has held about a third of every cohort since: **the largest grading family for new benchmarks in 2025 is programmatic**, at 29 of 80 (36%). Holistic LLM-judging peaked in 2024 (15 papers, 18%) and has receded since.

	proxy metric	exact match	programmatic	human	LLM judge	rubric + judge
L1 · seconds	12	54	14	6	2	
L2 · minutes	23	84	82	9	24	14
L3 · hours	1	7	24	1	1	7
L4 · days			1	2		1
L5 · weeks						

Figure 5: Task horizon $\times$ grading family, all 369 papers; hollow cells are empty. The mechanical families thin out with horizon but do not die: 7 exact-match and 24 programmatic papers survive at L3, and L4 is split 2 human, 1 rubric, 1 programmatic.

The genuinely new wave is rubric-scored judging, and **its slope is the story rather than its size**: 4 papers *total* before 2025 (FactScore (2023), SOTOPIA (2023), TIFA (2023) and Agent-as-a-Judge (2024)), then 15 of 80 (19%) in a single year — HealthBench, MultiChallenge, PaperBench and ToolEyes among them — and 3 of 14 in 2026 (HiLBench, MCP-Atlas and SWE Atlas). Set against 29 new programmatic benchmarks in the same year, that is a turn, not a takeover. We flag this explicitly because it is where a smaller corpus misleads: read on 139 papers, rubric judging looked like 2025’s largest grading family, and the fuller corpus falsifies that.

The wave structure has a single driver: each new capability regime produced outputs the previous grader could not score. Exact match broke on free-form generation; unit tests broke on anything that is not code; holistic judges broke on expert domains where “sounds good” diverges from “is correct.” The rubric turn is the field’s current answer — decompose expert judgment into criteria a model *can* check — and it is also a quiet confession that the thing we now want to measure no longer has an answer key. That the mechanical families still grade most new benchmarks says the field has not abandoned verifiability; it has partitioned, sending checkable work to harnesses and open-ended work to rubrics.

4.2 The verifiability frontier

Read Figure 5 diagonally and a boundary appears: **cheap grading gets rarer as horizons lengthen, and it survives only where the answer stays short**. Exact match owns L1 (54 of the 88 L1 papers) and thins to 7 of 41 at L3 — TAPE, LongICLBench, RULER, AgentDebug, BrowseComp, TRAIL and Who&When — none of which is a general-purpose answer key. They are short keys bolted onto long work: needle retrieval inside a very long context, search questions inverted until the answer is a verifiable string, and error localization in a recorded agent trace against a fixed label set. Programmatic grading peaks at L2–L3 (24 papers: unit tests, environment end-states) and nearly disappears at L4, where it survives in 1 paper — a game environment whose end state is still machine-readable. Nobody has built a test harness for “produce a competitive research paper.”

This is the survey’s central claim: **a benchmark can only pose tasks as long as it can afford to grade them**. Verification asymmetry — tasks that are hard to do but cheap to check — is the scarce resource of benchmark design, and the L4 row shows what happens when it runs out. GDPval pays experts for more than an hour of pairwise grading per item; PaperBench [Starace et al., 2025] amortizes thirty-three person-weeks of rubric authoring. The grading-cost ladder of Section 2 is not a menu of equally available options; at high horizons it is a budget constraint.

4.3 The horizon climb, and the empty top rung

Benchmarks are climbing the same ladder their subjects climb (Figure 6). In the corpus’s first two decades only 2 papers posed a task above L2 — TAPE (2019) and MLPerf Training (2020), and one of those, MLPerf, is a throughput harness rather than a capability eval. The L3+ share was 4–12% through 2023–24, then jumped to 28% in 2025 and reached 7 of 14 papers in 2026 (50%); the mean rung moved from 1.3 (≤ 2018) to 2.5 (2026). That tracks the

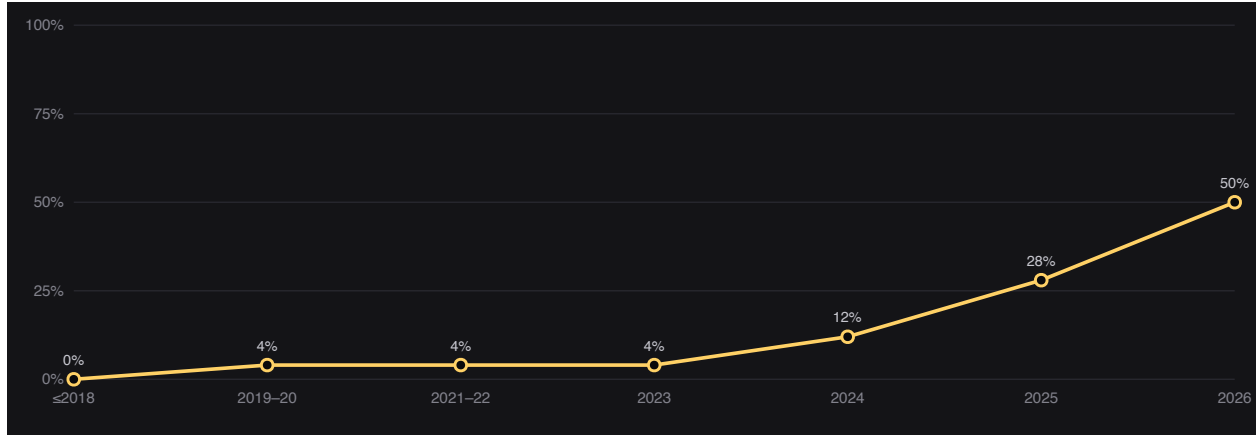


Figure 6: Share of each era’s new benchmark papers posing L3+ tasks (≥ 30 expert-minutes per instance).

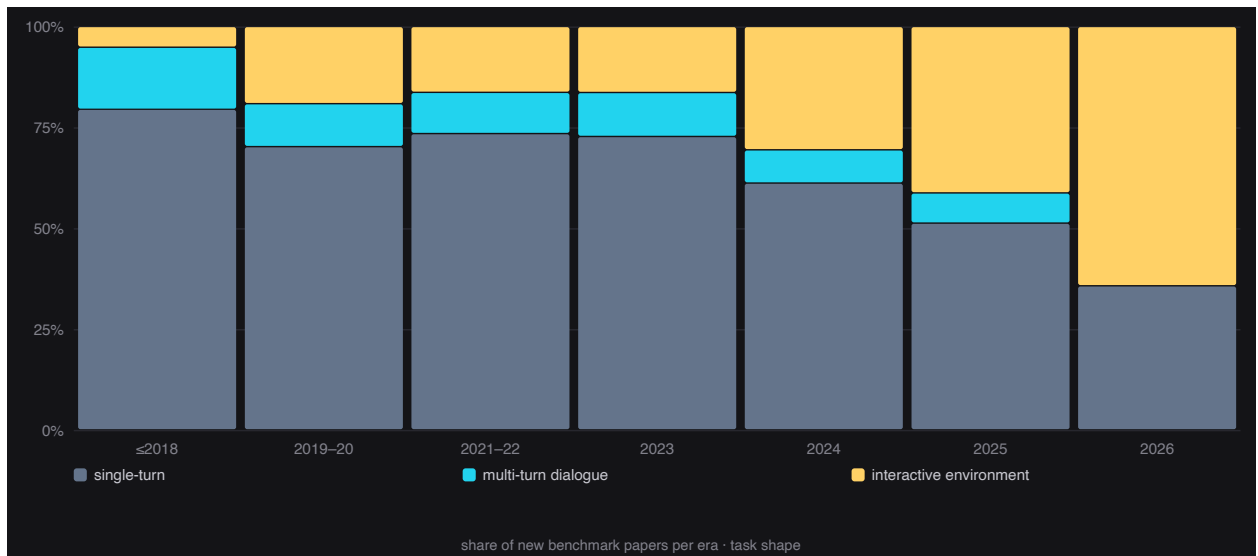


Figure 7: Task shape by era. Interactive environments: the model acts, the environment responds, repeat. Multi-turn: dialogue without an environment.

measured capability curve: as a model’s 50%-success horizon doubles roughly every seven months, benchmarks below the frontier stop discriminating and the field rebuilds one rung up.

But the climb is running out of ladder. L4 — day-to-week projects — contains 4 papers (BALROG, GDPval, Paper-Bench and RLI), all from 2025, and **L5 holds 0**: no benchmark in this corpus poses work beyond two expert-weeks. The binding constraints are the two graders left at that altitude (Section 4.2) and item economics — a thousand-item benchmark of week-long tasks is a thousand expert-weeks of ground truth. Whoever solves cheap verification of long-horizon work gets to write the first L5 benchmark, and probably the reward signal for the next training era with it.

4.4 The interactive turn is a return, and multi-turn dialogue is what got dropped

Interactive environments are not new here, and reading the recent climb as an invention is the mistake a smaller corpus invites. The first one is ALE (2013) [Bellemare et al., 2012], and the 2019–20 cohort is already 19% interactive (9 papers, Meta-World, Hanabi, RLBench and robosuite among them) — but that block evaluates *policies*, in robotics simulators and multi-agent games. The two generations do touch: 31 in-corpus citations run between them. They run through robotics and games, though — LIBERO [Liu et al., 2023a] and ManiSkill2 [Gu et al., 2023] citing RLBench [James et al., 2020] and Meta-World [Yu et al., 2019], BALROG [Paglieri et al., 2024] citing ALE — and

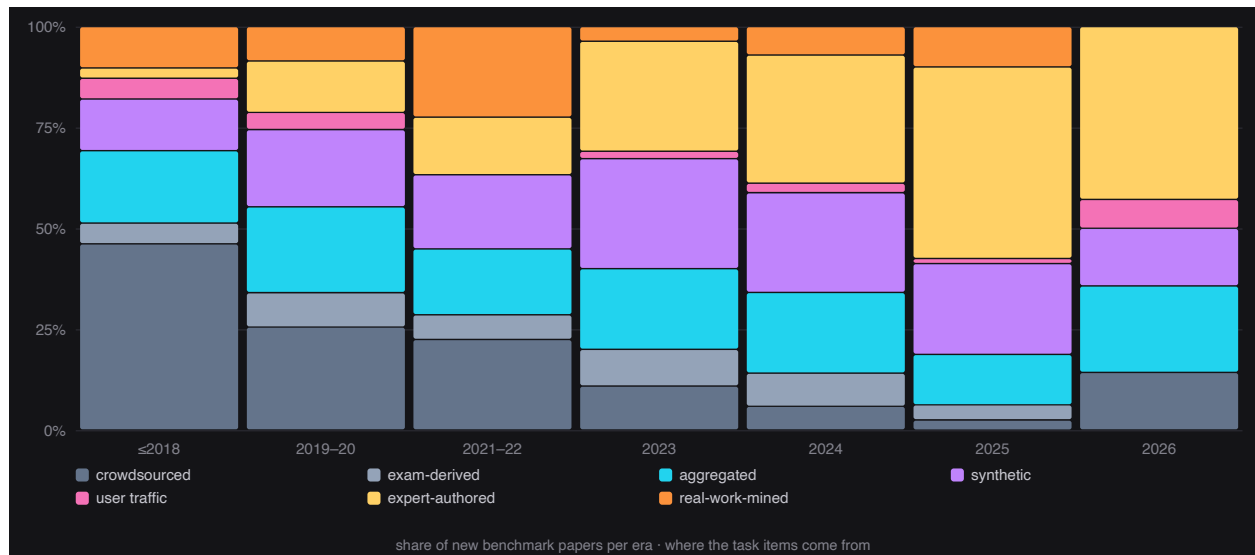


Figure 8: Provenance of task items by era. “Expert-authored” means items written or curated for the benchmark by recruited domain experts; “real-work-mined” means harvested from artifacts of real work (GitHub issues, freelance jobs, filed contracts).

not through the web-and-tool environments that drive the recent climb, which descend from WebShop [Yao et al., 2022] rather than from Gym.

The language-model era, in other words, did not inherit the form: the interactive share sits at 16% in 2021–22 and 16% in 2023, still mostly simulators and games, with WebShop the language-agent beachhead in 2022. Only then does it climb, on a different lineage: 31% in 2024, 41% in 2025 (33 of 80), 64% in 2026 (9 of 14). The static single-turn dataset was 74% of the corpus through 2020 and is 36% of 2026, already the minority format. The migration the agent surveys narrate is real; what the full timeline adds is that the field *re-derived* a form it had abandoned, from different ancestors.

The anomaly is what got dropped. **Multi-turn dialogue is 35 papers out of 369**, and its high-water mark is the *oldest* era in the corpus (2018 and earlier, 15%), when dialogue systems were the product and 11 pre-2021 papers evaluated conversation directly (DailyDialog, SMN, CMU DoG and PersonaChat). No era since has matched that share: 8% in 2024, 8% in 2025, 0 papers in 2026. The field did not skip multi-turn dialogue — it built the dialogue corpora, then walked away from them, jumping from one-shot prompts to autonomous tool loops and leaving the thing users actually do with these systems (talk, clarify, correct, change their minds mid-task) to a handful of instruments like MT-Bench [Zheng et al., 2023], MultiChallenge [Sirdeshmukh et al., 2025] and AdvancedIF [He et al., 2025]. Simulated users in the style of τ -bench [Yao et al., 2024] live inside a few agentic environments, but conversation as the *object* of evaluation is a shrinking share of the field, not a growing one.

4.5 Benchmarks stopped being scraped and started being commissioned

Through 2018, benchmarks were built from what could be collected cheaply: crowdworkers and scraped exams (20 of 39 papers, 51%). By 2025 crowdsourcing has collapsed to 2% and **48% of new benchmarks are expert-authored** — physicians writing HealthBench [Arora et al., 2025] criteria, olympiad medalists writing FrontierMath-class problems, industry professionals with a fourteen-year average tenure writing GDPval deliverables — and another 10% are mined from real work products (SWE-bench’s GitHub issues, SWE-Lancer’s [Miserendino et al., 2025] paid freelance jobs, RLI’s [Mazeika et al., 2025] real projects). The era of the annotation farm is over at the frontier.

The driver is the same as in Section 4.1: models passed the crowd. When the evaluated system outperforms a crowd-worker at the task, crowd-sourced ground truth stops being ground truth, item-writing becomes skilled labor, and benchmark construction industrializes — which is also why item counts fall as horizons rise. The economics of expertise, not dataset fashion, is what moved this chart.

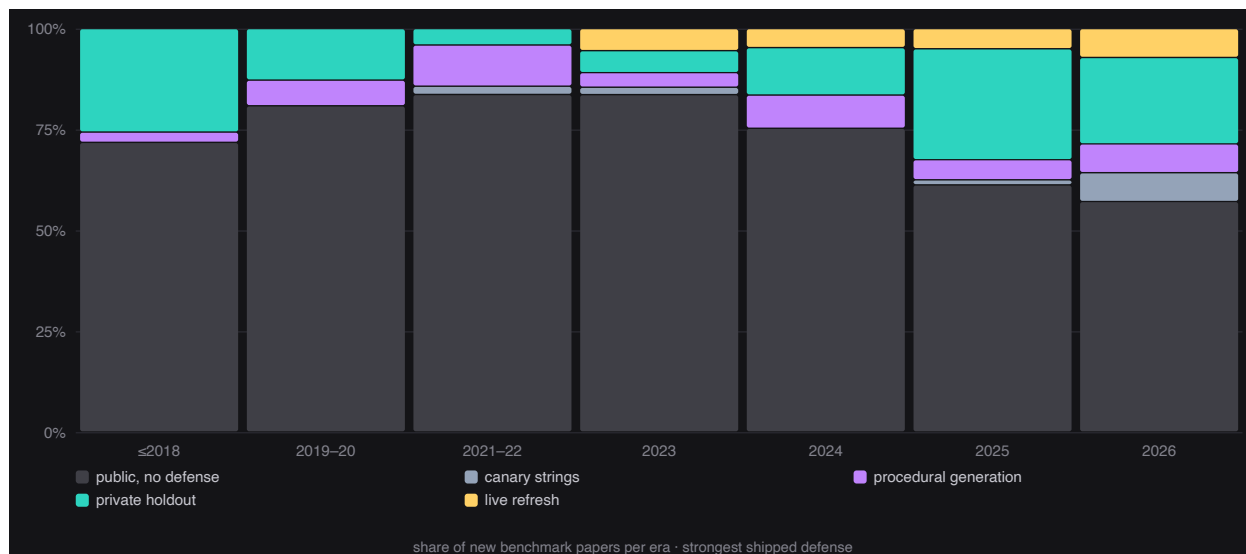


Figure 9: Strongest shipped contamination defense by era. “Private holdout” includes hidden test sets and evaluation servers; “live refresh” means post-cutoff or continuously regenerated items by design.

4.6 The contamination cycle

The instructive row is one of the oldest. The corpus’s earliest era already had the answer: 10 of 39 papers through 2018 (26%) shipped a hidden test set or an evaluation server — MS COCO, SQuAD, GLUE and HotpotQA among them — because a benchmark you could only query through a leaderboard was the norm when the test set was the scarce asset. The web-scrape training era then let it lapse: 41 of 49 papers in 2021–22 published everything (Codex, GSM8K, MATH and WebShop), leaving only 16% with any defense at all, and the contamination crisis of 2023–24 — GSM1k [Zhang et al., 2024] measured up to eight points of inflation on GSM8K — was the bill for that amnesia. From 2024 the defenses return and diversify: private holdouts (22 of 80 papers in 2025, 28%), live refresh (LiveCodeBench [Jain et al., 2024], SciPredict’s [Schwag et al., 2026] post-cutoff sourcing), procedural generation (Reasoning Gym [Stojanovski et al., 2025]), and canary strings.

The recovery is real and still partial, and the honest way to state it is as a level, not a crossing. The share of new benchmarks shipping *some* defense runs 28% (through 2018) → 19% → 16% → 16% → 25% → 39% → 43% (6 of 14 in 2026): a dip and a return, with the 2020s cohorts finally passing the era that invented the practice — but **no era in 26 years has ever crossed half**, peaking at 43%. The default benchmark, in every era of this corpus, ships public and undefended.

The defenses also work differently well. Of the 12 live-refresh benchmarks, **0 are saturated**; private holdouts saturate at 23%, slightly *above* the corpus base rate of 19%. A hidden test set stops leakage but not progress. Refresh-by-design is the only defense that also buys longevity, which is why the newest entries — SciPredict’s “no paper before March 2025,” UpBench’s [Yi et al., 2025] live freelance feed — build refresh into the sampling frame rather than bolting privacy onto a frozen set.

4.7 Horizon is destiny

Saturation is almost perfectly ordered by the ladder: **44% of L1 benchmarks are saturated, 12% of L2, and 2% of L3 and above** (Figure 10). Of the corpus’s 69 saturated benchmarks, 1 poses a task longer than half an expert-hour — TAPE (2019), a protein-modelling suite whose long tasks are graded against a numeric answer key. That exception is the rule restated: it is the one shape of long-horizon benchmark that can saturate, because it never had to pay for open-ended grading. Domain matters less than one might expect, and defense matters less than refresh (Section 4.6); what predicts whether a benchmark is still discriminating in 2026 is simply how much expert time its tasks embody.

That gives builders a clean actuarial rule — *expected lifespan is bought in human-minutes per item* — and it explains the corpus’s own shape. The L1 rows are the field’s history; the L2 rows are its present workhorses, 17% of them already closing; the L3–L4 rows, 73% of which are still open, are where evaluation currently lives. The ladder of Section 4 is not merely being climbed; it is being consumed from the bottom.

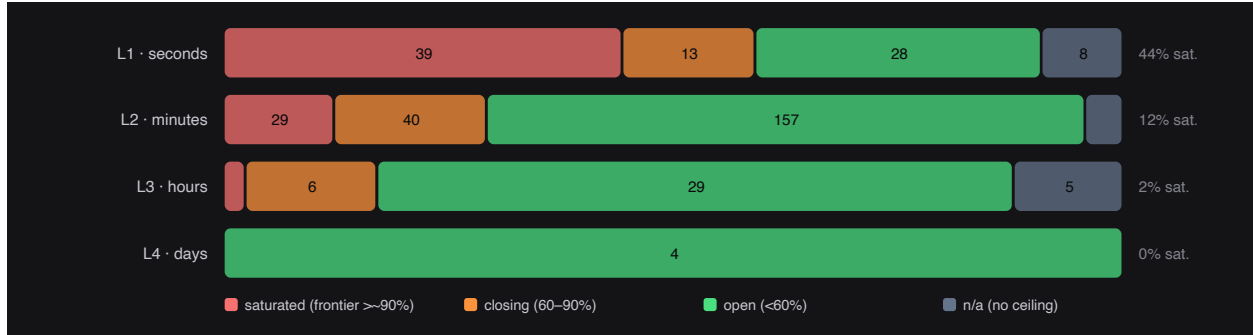


Figure 10: Saturation status by task horizon, judged from the frontier scores each explainer quotes (mid-2026). Counts inside bars.



Figure 11: First *corrective* descendant of the most-cited targets that have one: a child that audits its parent, de-saturates it, or hardens its grader. Extension relations (descends-from, responds-to, ports-to-new-domain) are excluded, and the gap between the two is the finding.

5 Cross-paper connections

The companion page reports nine connections mined by tracing the corpus’s internal citation graph (2013 edges among the 369 papers, plus 682 typed lineage relations recorded per paper during classification) against the full text of every explainer. We detail three here, chosen because each sharpens a finding above. Each is visible only across papers.

5.1 The audit shadow is not universal

Reading the lineage graph for *corrective* descent only turns the audit shadow from a slogan into a count (Figure 11). 48 such edges exist among the 369 papers, reaching 32 distinct targets, and the longest unaudited stretch in the corpus is ImageNet $\rightarrow$ ImageNet-C/P at 10 years. The wait has been shortening: the longest first-correction lag is 10 years for targets published through 2018, 5 for 2019–20 targets, 3 for 2021–22, and 2 for 2023 and later (medians 3, 4, 2.5, 1). Part of that is real and part of it is arithmetic — a 2024 benchmark *cannot* have a five-year-old audit yet — so the honest version of the claim is the one the early rows carry on their own: the pre-2019 cohort contains both a one-year correction and a ten-year one, and the recent cohorts contain no long waits at all.

The stronger claim, that every benchmark which mattered eventually acquired an audit, is false, and the graph is what falsifies it. **Of the corpus’s twenty most-cited benchmarks, 12 have a corrective descendant here.** The rest — SuperGLUE, WebShop, AgentBench and Gorilla among them — have plenty of descendants, but those descendants *extend* rather than *correct*: new domains, new modalities, harder instances of the same format. Correction and extension are different evolutionary pressures, and the agent-era gravity wells have so far attracted mostly the second. That is worth watching, because extension inherits a benchmark’s construct without ever testing it.

Where the shadow does fall, its mechanism has changed. Early corrections were third-party forensics — a Berkeley group rebuilding ImageNet’s test set [Recht et al., 2019] — but by 2024–25 they moved in-house and turned preemptive: OpenAI hardened its own SimpleQA [Wei et al., 2024] into BrowseComp [Wei et al., 2025], LMSYS distilled its own arena into Arena-Hard [Li et al., 2024], Scale extended its own MultiChallenge into an audio twin. Quality control became contemporaneous with publication. Read from the supply side, this is the same lifespan clock as Section 4.7: the shorter a benchmark’s discriminating life, the sooner its successor must already be in press.

5.2 The benchmark is the reward

The canonical benchmarks were never eval-only artifacts. GSM8K shipped as verifier-training infrastructure — a 7.5K training split plus a learned verifier, with the launch paper already documenting that verifier being reward-hacked. HumanEval [Chen et al., 2021] arrived inside the Codex training paper, its unit tests doubling as a sample-selection oracle; SWE-bench shipped 19K training instances and its own fine-tunes. When GSM1k later audited the ecosystem, its most telling overfitting case was contaminated through the *grader lineage* — a process-reward model leaking reasoning chains — with no test-set leak an n -gram checker could see.

Validity then had to be re-purchased, each time by moving the grader further from the training corpus: EvalPlus [Liu et al., 2023b] densified HumanEval’s tests roughly eightyfold; SWE-bench Pro [Deng et al., 2025] bought commercial code models could never have seen, and its 25-to-30-point public-to-commercial score collapse — same models, same harness — is the field’s cleanest single measurement of what training against a public grader family costs. By 2025 the flip is doctrine: MLGym’s [Nathani et al., 2025] complaint about prior benchmarks is that they *cannot* be trained on, and Reasoning Gym publishes its reward function outright. Goodhart does not vanish under procedural generation; it migrates into the grader’s blind spots.

5.3 Gravity selects portability

APPS [Hendrycks et al., 2021] and HumanEval are same-year, mutually citing, execution-graded code benchmarks; APPS was some sixty times larger, with twice the tests per problem. Yet HumanEval’s paper is the most-cited node in this corpus — 66 in-corpus citations, ahead of MMLU’s 63 and against APPS’s 20 — because what the field reused was the *measurement kit*: the unbiased $\text{pass}@k$ estimator plus 164 cheap, self-contained tasks scoring in the discriminative range. APPS’s grader vocabulary failed to travel (its lenient metric flattered; its strict metric floored at zero on the whole competition tier). The clincher: when EvalPlus proved HumanEval’s thin tests inflate $\text{pass}@k$ by up to about 29%, the field patched HumanEval in place rather than migrate to APPS — which already had denser tests.

The pattern repeats a generation down. SWE-bench Pro’s forkable Docker harness is already substrate for third-generation benchmarks, while SWE-Lancer’s more rigorous but unforkable grader — triple-verified end-to-end tests on one proprietary repository — has 0 coding descendants. Every benchmark that descends from it (GDPval, RLI and UpBench) took the *framing*, real-dollar difficulty, into the economic-rehearsal family instead, and the coding benchmarks that cite it do so to explain what they did differently. Benchmark genes travel through cheap, reproducible harnesses: the ecosystem structurally selects reproducibility over rigor.

6 Method and caveats

Corpus. The 369 papers are the atlas’s benchmarks survey tag, applied when each paper joined the atlas under the headline-contribution rubric of Section 2. The set over-represents what the atlas covers deeply — language-model, agent, and safety evaluation from 2023 onward, which is $55 + 85 + 80 + 14$ of the 369 — and samples the earlier generations rather than enumerating them, so era-level trends should be read as trends *within this kind of corpus*, not a census of all benchmarks ever built. The corpus is papers only; the 2026 model-card canon also includes non-paper instruments this survey does not cover (SWE-bench Verified, Aider Polyglot, AIME-as-eval, the ARC-AGI leaderboards, needle-in-a-haystack probes), an exclusion of form rather than a judgment that they do not matter.

Classification. Each paper was classified from its atlas explainer (and the paper itself where the explainer underdetermined a dimension) by a language-model pipeline, in two historical stages. The first 139 papers got one full-schema

pass followed by a fully independent second pass re-classifying the six judgment-heavy dimensions with different paper groupings; the two passes agreed on 94.8% of the 750 compared values, and all 39 disagreements, on 32 papers, were adjudicated by a third read of the source text, with saturation the noisiest dimension (16 of the 39 conflicts). The remaining 230 were classified in a later backfill, sharded into 12 batches of roughly 20 against the same schema, then merged, after which in-corpus citation in-degree was recomputed across the whole corpus and the lineage graph re-enriched — a shard can only see the papers inside it. Those two stages did not get equal scrutiny and the record carries the difference: 151 of 369 records are marked confidence: medium. Treat individual facet values — saturation above all — as softer than any aggregate here.

Tree placement. Kingdom and family addresses were assigned by the same protocol, judging role at birth from each paper’s own problem statement, with lineage restricted to relations between papers both in the corpus (682 of them, over 2013 citation edges among the 369). The independent second placement pass covered 78 papers of the original 139-paper corpus before rate-limiting and disagreed on 5 placements, each re-adjudicated against the source text. Two known lineage biases run in the same direction: a shard could not link to a parent sitting in a sibling shard (the post-merge enrichment pass exists to repair that, and did, but cannot prove it caught everything), and because a node’s date is its arXiv v1 month while its camera-ready cites later work, genuinely forward-pointing relations were dropped rather than recorded. The lineage graph therefore understates connectivity, in both cases arbitrarily with respect to the subject matter.

Connections. The connection stories were mined by lens-directed search over the corpus and its citation graph, then verified two ways: every quotation was checked against the explainer texts, and every citation-structure assertion (“X cites Y,” “X and Y share no edge”) was checked mechanically against the graph. Claims that survived only partially were corrected or dropped; the numbers inside the stories are the source papers’ own.

Known soft spots. Task-horizon rungs are estimates of the *typical* instance from the papers’ own task descriptions; only a few papers measure human time directly. Saturation reflects the frontier scores quoted in each explainer through mid-2026, and it ages. The recorded grading family is the headline metric only; many benchmarks ship secondary graders. Item counts are approximate. The domain vocabulary cannot name robotics, multi-agent reinforcement learning, or theory of mind, all of which the corpus contains, so any domain-sliced number inherits that squeeze. And one dimension a future pass should add is whether each benchmark’s grader was itself adversarially audited — the agentic-benchmark checklist’s 7-of-10 finding suggests that column would be mostly empty.

7 Conclusion

Placed in a single structure, 369 benchmarks stop looking like a scoreboard and start looking like an ecosystem governed by one economics. A benchmark can only pose tasks as long as it can afford to grade them, so the reach of evaluation is bounded by the cost of its grader; the verifiable, cheap-to-grade benchmarks are also the reward functions models train against; and a benchmark’s lifespan is bought in human-minutes per item, which is why the field is climbing the horizon ladder and consuming it from the bottom at the same time. Capability probes — the thing “benchmark” usually connotes — are already a minority of what the field builds each year (41% of 2025), the rest auditing, rehearsing, walling off, and manufacturing signal in a reflexive ecosystem a topic-based taxonomy cannot see. Two of this survey’s corrections to its own earlier, smaller version generalize past it: a long tail of pre-LLM work (86 papers, 23% of the corpus) turns several apparent arrivals into returns, and reading a lineage graph for *corrective* rather than merely descendant edges turns a universal claim about auditing into 12 of twenty. The corpus, every figure, and every statistic in this paper are generated from the live companion survey and regenerate deterministically as the corpus grows.

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